Wenbin Lu —— Curriculum Vitae

CONTACT INFORMATION

Address: Department of Astronomy
University of California Berkeley
501 Campbell Hall, # 3411
Berkeley, CA 94720, USA

Email: wenbinlu@berkeley.edu

https://wenbinlu.github.io

RESEARCH INTEREST

My research has been focused on understanding various high-energy transient phenomena, including *tidal disruption events*, *quasi-periodic eruptions*, *fast blue optical transients*, *fast radio bursts*, *fast X-ray transients*, *gamma-ray bursts*, *supernovae*, and *compact object mergers*. My overarching goal is to use them to probe stellar physics, radiation processes, accretion physics, and compact objects.

EMPLOYMENT

Assistant Professor, *University of California Berkeley*, California, USA	22-
Lyman Spitzer Fellow, *Princeton University*, New Jersey, USA	21-22
Burke Fellow, *California Institute of Technology*, California, USA	18-21

EDUCATION

Ph.D. in Astronomy, *University of Texas at Austin*, Texas, USA	13-18
B.S. in Physics, *Peking University*, Beijing, China	09-13

HONORS & AWARDS

Sloan Fellow	26
Scilog Fellow ("Early Science with LSST")	25, 26
Hellman Fellow ("Supernovae in Binaries")	24
Rose Hills Innovator ("Binary Stars at Galactic Centers")	23-24
Spitzer Postdoc Fellow	21-22
Burke Postdoc Fellow	18-21

PROFESSIONAL EXPERIENCES

Open-Access Book: Astrophysical Radiation (Zenodo link here)

Teaching

Graduate Course: AstroC207 – *Radiation* (22 Fall, 23 Fall, 25 Spring, 26 Spring)

Undergraduate Courses: Astro160 — *Stellar Physics* (24 Fall, 25 Fall, 26 Fall); Astro84 — *Astrophysical Origins of Chemical Elements* (23 Spring)

Mentoring

Graduate Students

Stephon K. Qian (Astro)	25-
Howard H.-T. Huang (Astro)	24-
Savannah Cary (Astro)	23-
Kishore Patra (Astro → UCSC postdoc)	22-24

Postdocs

Calvin Leung, Hubble Fellow	23-
Yuhan Yao, Miller Fellow (→ faculty at PKU)	23-
Payel Mukhopadhyay, N3AS Fellow (→ industry)	22-23

Undergraduate Students (* marks external or exchange students)

Jovan JohnPeter (Astro)	25-
Howard H.-T. Huang (→ UC Berkeley grad student)	22
Darby McCauley (Astro → UIUC grad student)	22-24
*Gauri Batra (Cornell → Stanford grad student)	22-23

Broader Services

Referee for *Nature*, *MNRAS*, *ApJ*, *ApJL*, *PRL*, *PRD*, *Space Science Reviews*, *JHEAP*, *Universe*	16-
Reviewer Panelist for proposals to NSF and NASA	
Reviewer for observing proposals to FAST telescope	
Reviewer for consolidated grant application to the Science & Technology Facilities Council (STFC) of the UK	
Chair for panel discussion in FRB2022 Workshop, Cornell University	22
LOC for FRB2021 Conference, University of Amsterdam	21

Involvement in Future Observing Facilities

UVEX (PI Harrison): Co-I and Deep Synoptic Surveys WG	22-
LS4 (PI Nugent): Multi-Messenger WG	23-

Colloquia

Carnegie Observatory Colloquium Series	05/2026
Astronomy Colloquium, Indiana University	04/2026
Astronomy Seminar, Institute of Science and Technology Austria	03/2026
Munich Joint Astronomy Colloquium, Max-Planck Institute	03/2026
Astronomy Colloquium, University of California Santa Cruz	04/2025
Astronomy Colloquium, University of Illinois Urbana-Champaign	11/2024
Physics Colloquium, University of Nevada Las Vegas	04/2024
Physics Colloquium, California State University Sacramento	09/2023
Astronomy Colloquium, Tsinghua University	03/2021
Astronomy Colloquium, University of California Berkeley	02/2021
Institute of Theory and Computation Colloquium Series, Harvard	10/2020
McGill Space Institute Astrophysics Seminar	09/2020
Carnegie Observatories Colloquium	02/2020
Astronomy Colloquium, California Institute of Technology	01/2020
Physics and Astronomy Colloquium, University of Nevada at Las Vegas	11/2018
Kavli Institute for Astronomy and Astrophysics Colloquium, Peking University	09/2018
Black Hole Initiative Colloquium, Harvard University	04/2017

Seminars

TAPIR Seminar, Caltech, USA	03/2024
Online Seminar, Max-Planck-Institute for gravitational physics, Germany	05/2023
X-ray Binary Seminar, University of Southampton, UK	03/2023
THEA Seminar, Columbia University, USA	05/2022
Astroplasma Seminar, Princeton University, USA	02/2022
High-Energy Astrophysics Seminar, Hebrew University of Jerusalem, Israel	12/2021
Brown Bag Lunch, Massachusetts Institute of Technology, USA	05/2021
Theoretical Astrophysics Seminar, University of Florida, USA	10/2020

Astroplasmas Seminar, Princeton University, USA 10/2020

Astrophysics Lunch, Cornell University, USA 09/2020

(Blackboard) Carnegie Theory Talks, Pasadena, USA 07/2020

TAPIR Seminar, Caltech, USA 10/2018

(Blackboard) Carnegie Theory Talks, Pasadena, USA 09/2018

Astroplasmas Seminar, Princeton University, USA 12/2017

Astronomy Tea Talk, Caltech, USA 10/2017

Transient Lunch, UC Santa Cruz, USA 09/2017

Theoretical Astrophysics Center Seminar, UC Berkeley, USA 09/2017

(Blackboard) Institute of Theory and Computation, Harvard University, USA 04/2017

Lunch Talk, University of Kentucky, USA 10/2016

Astronomy Seminar, University of Science and Technology of China, China 05/2016

Lunch Talk, KIAA/Peking University, China 05/2016

Lunch Talk, University of Nevada at Las Vegas, USA 05/2015

Invited Conference Talks

Formation of Ultra-Long Period Pulsars 10/2025

FRB Workshop, Zhejiang University, China

Broad-Brush Model for Quasi-Periodic Eruptions 04/2024

TDE Workshop, UC Santa Barbara, USA

Late-time Accretion in Neutron Star Mergers 12/2023

Gravitational Wave workshop, Tokyo University, Japan

Origin of Quasi-Period Eruptions 12/2023

Texas Symposium, T.D. Lee Institute, China

Implications of the FRB source in the M81 Globular Cluster 05/2023

FRB Workshop, University of Science and Technology of China, Hefei, China

Understanding the Fast Radio Bursts in our Local Universe 10/2022

FRB2022 Workshop, Cornell University, Ithaca, NY

Towards Understanding of Fast Radio Bursts 02/2021

FRB workshop, Yukawa Institute for Theoretical Physics, Kyoto University

General Constraints on the Emission Mechanisms of Fast Radio Bursts 02/2020

FRB workshop, CCA Flatiron Institute, New York

Implications of Stream Self-Crossing in Tidal Disruption Events 01/2020
TDE workshop, Yukawa Institute for Theoretical Physics, Kyoto University

Accretion Disk Evolution in Binary Neutron Star Mergers 09/2019
ZTF Theory Network Meeting, San Luis Obispo, USA

Statistical and Polarization Properties of Fast Radio Bursts 09/2019
Toronto FRB Day, CITA and University of Toronto

Energetics and Polarization Properties of Fast Radio Bursts 01/2019
T. D. Lee Institute mini-workshop, Shanghai, China

Understanding the Polarization of Fast Radio Bursts 07/2018
ZTF Theory Network Meeting, Santa Barbara, USA

Contributed Conference Talks

Origin of Quasi-Periodic Eruptions 03/2023
Aspen Winter Conference, Aspen Center for Physics, USA

Aftermath of white dwarf tidal capture 01/2022
Aspen Winter Conference, Aspen Center for Physics, USA

Implications of a rapidly varying FRB in a globular cluster of M81 08/2021
FRB2021 Conference, University of Amsterdam, the Netherlands

A Unified Picture of Galactic and Cosmological Fast Radio Bursts 07/2020
FRB2020 Conference, West Virginia University, USA

The Radiation Mechanism of Fast Radio Bursts 12/2017
Deciphering the Violent Universe, Playa del Carmen, Mexico

Public Talks

The Unluckiest Stars at the Center of Galaxies 10/2025
Astro Night at Campbell Hall, Berkeley, CA

How Gamma-Ray Bursts are Connected to Human Life 02/2023
Chabot Space and Science Center, Oakland, CA

Endless hunt for black holes 04/2021
Caltech Stargazing Lecture Series

Stories of black holes tearing apart stars at galactic centers 02/2017
McDonald Observatory & DoA Board of Visitors Meeting, UT Austin

General relativity and black holes 04/2016
Planetary Organization for Space Science and Exploration in Jackson School of Geosciences, UT Austin

PUBLICATION LIST ADS link to all my publications Google Scholar link

Published (h**-index 37**, m-index 3.1, total citations 4700+)

98. Sfaradi, I., Margutti, R., Chornock, R., A. J., N., Wiston, E., et al. **(Lu, W.)**, A 14-yr-old Mystery: The Peculiar Case of the Engine-driven SN 2012ap, *ApJL*, 1005, 1, (2026)(jul) PDF

97. Rose, K., Pritchard, J., Murphy, T., Driessen, L., Kaplan, D., et al. **(Lu, W.)**, Periodic radio and X-ray emission from an accreting white dwarf binary, *Nature Astronomy*, (2026)(jun) PDF

96. Li, W., Christy, C., Alexander, K., Sfaradi, I., Huang, X., et al. **(Lu, W.)**, VLA Observations Confirm AT 2023mfm as an Off-nuclear Tidal Disruption Event, *Research Notes of the American Astronomical Society*, 10, 6, (2026)(jun) PDF

95. Yao, Y., Chornock, R., Mummery, A., Margutti, R., Gilfanov, M., et al. **(Lu, W.)**, AT2024lhc and AT2024kmq in the landscape of featureless tidal disruption events, *MNRAS*, 549, 2, (2026)(jun) PDF

94. Zheng, J., **Lu, W.**, Fast X-Ray Transients Produced by Off-axis Jet Cocoons from Long Gamma-Ray Bursts, *ApJL*, 1003, 1, (2026)(may) PDF

93. Huang, H., **Lu, W.**, Formation of Close Binaries through Massive Black Hole Perturbations and Chaotic Tides, *The Open Journal of Astrophysics*, 9, (2026)(may) PDF

92. Scherbak, P., **Lu, W.**, Fuller, J., Radiatively Cooled Binary Mass Transfer: Flow Structure, Luminosities, and L2 Outflows Across Mass Transfer Rates, *pasp*, 138, 3, (2026)(mar) PDF

91. LeBaron, N., Margutti, R., Chornock, R., A. J., N., Aspegren, O., et al. **(Lu, W.)**, The Most Luminous Known Fast Blue Optical Transient AT 2024wpp: Unprecedented Evolution and Properties in the Ultraviolet to the Near-infrared, *ApJL*, 997, 1, (2026)(jan) PDF

90. Cary, S., **Lu, W.**, Leung, C., Wong, T., Accretion from a Shock-inflated Companion: Spinning Down Neutron Stars to Hour-long Periods, *ApJ*, 996, 2, (2026)(jan) PDF

89. Somalwar, J., Ravi, V., Margutti, R., Chornock, R., Natarajan, P., et al. **(Lu, W.)**, A Luminous and Hot Infrared through X-Ray Transient at a 5 kpc Offset from a Dwarf Galaxy, *ApJ*, 995, 2, (2025)(dec) PDF

88. Nayana, A., Margutti, R., Wiston, E., Laskar, T., Migliori, G., et al. **(Lu, W.)**, The Most Luminous Known Fast Blue Optical Transient AT 2024wpp: Unprecedented Evolution and Properties in the X-Rays and Radio, *ApJL*, 993, 1, (2025)(nov) PDF

87. Yao, Y., Alexander, K., **Lu, W.**, Somalwar, J., Ravi, V., et al., Optically Overluminous Tidal Disruption Events: Outflow Properties and Implications for Extremely Relativistic Disruptions, *ApJ*, 993, 2, (2025)(nov) PDF

86. Burn, M., Goodwin, A., Anderson, G., Miller-Jones, J., Cendes, Y., et al. **(Lu, W.)**, The 6 yr Radio Light Curve of the Tidal Disruption Event AT2019azh, *ApJ*, 993, 2, (2025)(nov) PDF

85. Sfaradi, I., Margutti, R., Chornock, R., Alexander, K., Metzger, B., et al. **(Lu, W.)**, The First Radio-bright Off-nuclear Tidal Disruption Event AT 2024tvd Reveals the Fastest-evolving Double-peaked Radio Emission, *ApJL*, 992, 2, (2025)(oct) PDF

84. Scherbak, P., **Lu, W.**, Fuller, J., Rapid Binary Mass Transfer: Circumbinary Outflows and Angular Momentum Losses, *ApJ*, 990, 2, (2025)(sep) PDF

83. Miller, A., Abrams, N., Aldering, G., Anand, S., Angus, C., et al. **(Lu, W.)**, The La Silla Schmidt Southern Survey, *pasp*, 137, 9, (2025)(sep) PDF

82. Masterson, M., De, K., Panagiotou, C., Kara, E., **Lu, W.**, et al., JWST's First View of Tidal Disruption Events: Compact, Accretion-driven Emission Lines and Strong Silicate Emission in an Infrared-selected Sample, *ApJL*, 988, 2, (2025)(aug) PDF

81. Feng, Y., Zhang, Y., Xie, J., Yang, Y., Qu, Y., et al. **(Lu, W.)**, Multi-year polarimetric monitoring of four CHIME-discovered repeating fast radio bursts with FAST, *Science China Physics, Mechanics, and Astronomy*, 68, 8, (2025)(aug) PDF

80. Yao, Y., Chornock, R., Ward, C., Hammerstein, E., Sfaradi, I., et al. **(Lu, W.)**, A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd, *ApJL*, 985, 2, (2025)(jun) PDF

79. Tsuna, D., **Lu, W.**, Stellar Tidal Disruptions by Newborn Neutron Stars or Black Holes: A Mechanism for Hydrogen-poor (Super)luminous Supernovae and Fast Blue Optical Transients, *ApJ*, 986, 1, (2025)(jun) PDF

78. Somalwar, J., Ravi, V., Yao, Y., Guolo, M., Graham, M., et al. **(Lu, W.)**, The First Systematically Identified Repeating Partial Tidal Disruption Event, *ApJ*, 985, 2, (2025)(jun) PDF

77. Wharton, R., Majid, W., Sherman, M., Connor, L., Kocz, J., et al. **(Lu, W.)**, High-frequency Fast Radio Burst Search of Nearby Star-forming Galaxies M77 and M82, *ApJ*, 984, 2, (2025)(may) PDF

76. Zheng, J., Zhu, J., **Lu, W.**, Zhang, B., EP240414a: Off-axis View of a Jet-cocoon System from an Expanded Progenitor Star, *ApJ*, 985, 1, (2025)(may) PDF

75. Earl, N., French, K., Ramirez-Ruiz, E., Auchettl, K., Raimundo, S., et al. **(Lu, W.)**, AT 2020nov: Evidence for Disk Reprocessing in a Rare Tidal Disruption Event, *ApJ*, 983, 1, (2025)(apr) PDF

74. Somalwar, J., Ravi, V., **Lu, W.**, VLASS Tidal Disruption Events with Optical Flares. II. Discovery of Two TDEs with Intermediate Width Balmer Emission Lines and Connections to the Ambiguous Extreme Coronal Line Emitters, *ApJ*, 983, 2, (2025)(apr) PDF

73. Goodwin, A., Mummery, A., Laskar, T., Alexander, K., Anderson, G., et al. **(Lu, W.)**, A Second Radio Flare from the Tidal Disruption Event AT2020vwl: A Delayed Outflow Ejection?, *ApJ*, 981, 2, (2025)(mar) PDF

72. Zhuang, J., Shen, R., Mou, G., **Lu, W.**, Interaction of an Outflow with Surrounding Gaseous Clouds as the Origin of Late-time Radio Flares in Tidal Disruption Events, *ApJ*, 979, 2, (2025)(feb) PDF

71. Yao, P., Quataert, E., Jiang, Y., **Lu, W.**, White, C., et al., Star−Disk Collisions: Implications for Quasi-periodic Eruptions and Other Transients near Supermassive Black Holes, *ApJ*, 978, 1, (2025)(jan) PDF

70. Yao, Y., Guolo, M., Tombesi, F., Li, R., Gezari, S., et al. **(Lu, W.)**, Subrelativistic Outflow and Hours-timescale Large-amplitude X-Ray Dips during Super-Eddington Accretion onto a Low-mass Massive Black Hole in the Tidal Disruption Event AT2022lri, *ApJ*, 976, 1, (2024)(nov) PDF

69. Orr, M., Burkhart, B., **Lu, W.**, Ponnada, S., Hummels, C., et al., Objects May Be Closer than They Appear: Significant Host Galaxy Dispersion Measures of Fast Radio Bursts in Zoom-in Simulations, *ApJL*, 972, 2, (2024)(sep) PDF

68. **Lu, W.**, Matsumoto, T., Matzner, C., Misaligned precessing jets are choked by the accretion disc wind, *MNRAS*, 533, 1, (2024)(sep) PDF

67. Patra, K., **Lu, W.**, Ma, Y., Quataert, E., Miniutti, G., et al., Constraints on the narrow-line region of the X-ray quasi-periodic eruption source GSN 069, *MNRAS*, 530, 4, (2024)(jun) PDF

66. Yao, Y., **Lu, W.**, Harrison, F., Kulkarni, S., Gezari, S., et al., The On-axis Jetted Tidal Disruption Event AT2022cmc: X-Ray Observations and Broadband Spectral Modeling, *ApJ*, 965, 1, (2024)(apr) PDF

65. Kirsten, F., Ould-Boukattine, O., Herrmann, W., Gawroński, M., Hessels, J., et al. **(Lu, W.)**, A link between repeating and non-repeating fast radio bursts through their energy distributions, *Nature Astronomy*, 8, (2024)(mar) PDF

64. Sheikh, S., Farah, W., Pollak, A., Siemion, A., Chamma, M., et al. **(Lu, W.)**, Characterization of the repeating FRB 20220912A with the Allen Telescope Array, *MNRAS*, 527, 4, (2024)(feb) PDF

63. Huang, H., **Lu, W.**, Tidal disruption rate suppression by the event horizon of spinning black holes, *MNRAS*, 527, 2, (2024)(jan) PDF

62. **Lu, W.**, Quataert, E., Quasi-periodic eruptions from mildly eccentric unstable mass transfer in galactic nuclei, *MNRAS*, 524, 4, (2023)(oct) PDF

61. Ryder, S., Bannister, K., Bhandari, S., Deller, A., Ekers, R., et al. **(Lu, W.)**, A luminous fast radio burst that probes the Universe at redshift 1, *Science*, 382, 6668, (2023)(oct) PDF

60. Yao, Y., Ravi, V., Gezari, S., van Velzen, S., **Lu, W.**, et al., Tidal Disruption Event Demographics with the Zwicky Transient Facility: Volumetric Rates, Luminosity Function, and Implications for the Local Black Hole Mass Function, *ApJL*, 955, 1, (2023)(sep) PDF

59. **Lu, W.**, Quataert, E., Late-time accretion in neutron star mergers: Implications for short gamma-ray bursts and kilonovae, *MNRAS*, 522, 4, (2023)(jul) PDF

58. Batra, G., **Lu, W.**, Bonnerot, C., Phinney, E., General relativistic stream crossing in tidal disruption events, *MNRAS*, 520, 4, (2023)(apr) PDF

57. Wang, B., Xu, H., Jiang, J., Xu, J., Niu, J., et al. **(Lu, W.)**, Atlas of dynamic spectra of fast radio burst FRB 20201124A, *Chinese Physics B*, 32, 2, (2023)(feb) PDF

56. **Lu, W.**, Fuller, J., Quataert, E., Bonnerot, C., On rapid binary mass transfer - I. Physical model, *MNRAS*, 519, 1, (2023)(feb) PDF

55. Kremer, K., Li, D., **Lu, W.**, Piro, A., Zhang, B., et al., Prospects for Detecting Fast Radio Bursts in the Globular Clusters of Nearby Galaxies, *ApJ*, 944, 1, (2023)(feb) PDF

54. Andreoni, I., Coughlin, M., Perley, D., Yao, Y., **Lu, W.**, et al., A very luminous jet from the disruption of a star by a massive black hole, *Nature*, 612, 7940, (2022)(dec) PDF

53. **Lu, W., Accretion Disk Evolution in Tidal Disruption Events, *Handbook of X-ray and Gamma-ray Astrophysics*, (2022)() PDF

52. Andreoni, I., **Lu, W.**, Grefenstette, B., Kasliwal, M., Yan, L., et al., Hard X-Ray Observations of the Hydrogen-poor Superluminous Supernova SN 2018hti with NuSTAR, *ApJL*, 941, 1, (2022)(dec) PDF

51. Mooley, K., Anderson, J., **Lu, W.**, Optical superluminal motion measurement in the neutron-star merger GW170817, *Nature*, 610, 7931, (2022)(oct) PDF

50. Kumar, P., Gill, R., **Lu, W.**, Propagation of Alfvén waves in the charge starvation regime, *MNRAS*, 516, 2, (2022)(oct) PDF

49. Yao, Y., **Lu, W.**, Guolo, M., Pasham, D., Gezari, S., et al., The Tidal Disruption Event AT2021ehb: Evidence of Relativistic Disk Reflection, and Rapid Evolution of the Disk-Corona System, *ApJ*, 937, 1, (2022)(sep) PDF

48. Patra, K., **Lu, W.**, Brink, T., Yang, Y., Filippenko, A., et al., Spectropolarimetry of the tidal disruption event AT 2019qiz: a quasi-spherical reprocessing layer, *MNRAS*, 515, 1, (2022)(sep) PDF

47. Xu, H., Niu, J., Chen, P., Lee, K., Zhu, W., et al. **(Lu, W.)**, A fast radio burst source at a complex magnetized site in a barred galaxy, *Nature*, 609, 7928, (2022)(sep) PDF

46. Kremer, K., Lombardi, J., **Lu, W.**, Piro, A., Rasio, F., et al., Hydrodynamics of Collisions and Close Encounters between Stellar Black Holes and Main-sequence Stars, *ApJ*, 933, 2, (2022)(jul) PDF

45. Bonnerot, C., Pessah, M., **Lu, W.**, From Pericenter and Back: Full Debris Stream Evolution in Tidal Disruption Events, *ApJL*, 931, 1, (2022)(may) PDF

44. Somalwar, J., Ravi, V., Dong, D., Graham, M., Hallinan, G., et al. **(Lu, W.)**, The Nascent Milliquasar VT J154843.06+220812.6: Tidal Disruption Event or Extreme Accretion State Change?, *ApJ*, 929, 2, (2022)(apr) PDF

43. Fuller, J., **Lu, W.**, The spins of compact objects born from helium stars in binary systems, *MNRAS*, 511, 3, (2022)(apr) PDF

42. Yang, Y., **Lu, W.**, Feng, Y., Zhang, B., Li, D., et al., Temporal Scattering, Depolarization, and Persistent Radio Emission from Magnetized Inhomogeneous Environments near Repeating Fast Radio Burst Sources, *ApJL*, 928, 2, (2022)(apr) PDF

41. Bonnerot, C., **Lu, W.**, The nozzle shock in tidal disruption events, *MNRAS*, 511, 2, (2022)(apr) PDF

40. Feng, Y., Li, D., Yang, Y., Zhang, Y., Zhu, W., et al. **(Lu, W.)**, Frequency-dependent polarization of repeating fast radio bursts—implications for their origin, *Science*, 375, 6586, (2022)(mar) PDF

39. **Lu, W.**, Beniamini, P., Kumar, P., Implications of a rapidly varying FRB in a globular cluster of M81, *MNRAS*, 510, 2, (2022)(feb) PDF

38. Makhathini, S., Mooley, K., Brightman, M., Hotokezaka, K., Nayana, A., et al. **(Lu, W.)**, The Panchromatic Afterglow of GW170817: The Full Uniform Data Set, Modeling, Comparison with Previous Results, and Implications, *ApJ*, 922, 2, (2021)(dec) PDF

37. **Lu, W.**, McKee, C., Mooley, K., Infrared dust echoes from neutron star mergers, *MNRAS*, 507, 3, (2021)(nov) PDF

36. Bij, A., Lin, H., Li, D., van Kerkwijk, M., Pen, U., et al. **(Lu, W.)**, Kinematics of Crab Giant Pulses, *ApJ*, 920, 1, (2021)(oct) PDF

35. Beniamini, P., **Lu, W.**, Survival Times of Supramassive Neutron Stars Resulting from Binary Neutron Star Mergers, *ApJ*, 920, 2, (2021)(oct) PDF

34. Connor, L., Shila, K., Kulkarni, S., Flygare, J., Hallinan, G., et al. **(Lu, W.)**, Galactic Radio Explorer: An All-sky Monitor for Bright Radio Bursts, *pasp*, 133, 1025, (2021)(jul) PDF

33. Bonnerot, C., **Lu, W.**, Hopkins, P., First light from tidal disruption events, *MNRAS*, 504, 4, (2021)(jul) PDF

32. **Lu, W.**, Fuller, J., Raveh, Y., Perets, H., Li, T., et al., The former companion of hyper-velocity star S5-HVS1, *MNRAS*, 503, 1, (2021)(may) PDF

31. Kremer, K., **Lu, W.**, Piro, A., Chatterjee, S., Rasio, F., et al., Fast Optical Transients from Stellar-mass Black Hole Tidal Disruption Events in Young Star Clusters, *ApJ*, 911, 2, (2021)(apr) PDF

30. **Lu, W.**, Beniamini, P., Bonnerot, C., On the formation of GW190814, *MNRAS*, 500, 2, (2021)(jan) PDF

29. Kool, E., Reynolds, T., Mattila, S., Kankare, E., Pérez-Torres, M., et al. **(Lu, W.)**, AT 2017gbl: a dust obscured TDE candidate in a luminous infrared galaxy, *MNRAS*, 498, 2, (2020)(oct) PDF

28. **Lu, W.**, Piro, A., Waxman, E., Implications of Canadian Hydrogen Intensity Mapping Experiment repeating fast radio bursts, *MNRAS*, 498, 2, (2020)(oct) PDF

27. **Lu, W.**, Kumar, P., Zhang, B., A unified picture of Galactic and cosmological fast radio bursts, *MNRAS*, 498, 1, (2020)(oct) PDF

26. De Colle, F., **Lu, W.**, Jets from Tidal Disruption Events, *New Astronomy Reviews*, 89, (2020)(sep) PDF

25. **Lu, W.**, Phinney, E., Imprint of local environment on fast radio burst observations, *MNRAS*, 496, 3, (2020)(aug) PDF

24. Chen, G., Ravi, V., **Lu, W.**, The Multiwavelength Counterparts of Fast Radio Bursts, *ApJ*, 897, 2, (2020)(jul) PDF

23. Andreoni, I., **Lu, W.**, Smith, R., Masci, F., Bellm, E., et al., Zwicky Transient Facility Constraints on the Optical Emission from the Nearby Repeating FRB 180916.J0158+65, *ApJL*, 896, 1, (2020)(jun) PDF

22. Bonnerot, C., **Lu, W.**, Simulating disc formation in tidal disruption events, *MNRAS*, 495, 1, (2020)(jun) PDF

21. Piro, A., **Lu, W.**, Wind-reprocessed Transients, *ApJ*, 894, 1, (2020)(may) PDF

20. Kumar, P., **Lu, W.**, Radiation forces constrain the FRB mechanism, *MNRAS*, 494, 1, (2020)(may) PDF

19. **Lu, W.**, Bonnerot, C., Self-intersection of the fallback stream in tidal disruption events, *MNRAS*, 492, 1, (2020)(feb) PDF

18. **Lu, W.**, Piro, A., Implications from ASKAP Fast Radio Burst Statistics, *ApJ*, 883, 1, (2019)(sep) PDF

17. Kremer, K., **Lu, W.**, Rodriguez, C., Lachat, M., Rasio, F., et al., Tidal Disruptions of Stars by Black Hole Remnants in Dense Star Clusters, *ApJ*, 881, 1, (2019)(aug) PDF

16. **Lu, W.**, Kumar, P., The maximum luminosity of fast radio bursts, *MNRAS*, 483, 1, (2019)(feb) PDF

15. **Lu, W.**, Kumar, P., Narayan, R., Fast radio burst source properties from polarization measurements, *MNRAS*, 483, 1, (2019)(feb) PDF

14. **Lu, W.**, Kumar, P., On the Missing Energy Puzzle of Tidal Disruption Events, *ApJ*, 865, 2, (2018)(oct) PDF

13. De Colle, F., **Lu, W.**, Kumar, P., Ramirez-Ruiz, E., Smoot, G., et al., Thermal and non-thermal emission from the cocoon of a gamma-ray burst jet, *MNRAS*, 478, 4, (2018)(aug) PDF

12. **Lu, W.**, Kumar, P., On the radiation mechanism of repeating fast radio bursts, *MNRAS*, 477, 2, (2018)(jun) PDF

11. Carballo-Rubio, R., Kumar, P., **Lu, W.**, Seeking observational evidence for the formation of trapping horizons in astrophysical black holes, *Phys. Rev. D*, 97, 12, (2018)(jun) PDF

10. Bhattacharya, M., **Lu, W.**, Kumar, P., Santana, R., Monte Carlo Simulations of Photospheric Emission in Relativistic Outflows, *ApJ*, 852, 1, (2018)(jan) PDF

9. **Lu, W.**, Krolik, J., Crumley, P., Kumar, P., Radiative interaction between the relativistic jet and optically thick envelope in tidal disruption events, *MNRAS*, 471, 1, (2017)(oct) PDF

8. Dai, L., **Lu, W.**, Probing Motion of Fast Radio Burst Sources by Timing Strongly Lensed Repeaters, *ApJ*, 847, 1, (2017)(sep) PDF

7. Kumar, P., **Lu, W.**, Bhattacharya, M., Fast radio burst source properties and curvature radiation model, *MNRAS*, 468, 3, (2017)(jul) PDF

6. **Lu, W.**, Kumar, P., Narayan, R., Stellar disruption events support the existence of the black hole event horizon, *MNRAS*, 468, 1, (2017)(jun) PDF

5. **Lu, W.**, Kumar, P., A universal EDF for repeating fast radio bursts?, *MNRAS*, 461, 1, (2016)(sep) PDF

4. Crumley, P., **Lu, W.**, Santana, R., Hernández, R., Kumar, P., et al., Swift J1644+57: an ideal test bed of radiation mechanisms in a relativistic super-Eddington jet, *MNRAS*, 460, 1, (2016)(jul) PDF

3. **Lu, W.**, Kumar, P., Evans, N., Infrared emission from tidal disruption events - probing the pc-scale dust content around galactic nuclei, *MNRAS*, 458, 1, (2016)(may) PDF

2. **Lu, W.**, Kumar, P., External inverse-Compton emission from jetted tidal disruption events, *MNRAS*, 458, 1, (2016)(may) PDF

1. **Lu, W.**, Kumar, P., Smoot, G., Probing massive stars around gamma-ray burst progenitors, *MNRAS*, 453, 2, (2015)(oct) PDF

Submitted

8. **Lu, W.**, Piro, A., Implications of the UV/optical Plateau of AT2018cow, *submitted*, arXiv: 2607.15464, (2026)(jul) PDF

7. McCauley, D., Parsons, A., Liu, W., **Lu, W.**, Wright, D., et al., Long-Integration Magnetar Burst Observatory (LIMBO): Instrument Summary and Early FRB Rate Constraints, *submitted*, arXiv: 2603.05603, (2026)(mar) PDF

6. Wang, Y., Wang, T., Huang, S., Zhu, J., Jiang, N., et al. **(Lu, W.)**, A Star's Death by a Thousand Cuts: The Runaway Periodic Eruptions of AT2023uqm, *submitted*, arXiv: 2510.26561, (2025)(oct) PDF

5. Tsuna, D., Fuller, J., **Lu, W.**, Fates of Rotating Supergiants from Stellar Mergers and the Landscape of Transients upon Core-collapse, *submitted*, arXiv: 2508.21116, (2025)(aug) PDF

4. **Lu, W.**, Cary, S., Tsuna, D., Accretion from a shock-inflated companion: double-peaked supernova lightcurve with periodic modulations, *submitted*, arXiv: 2507.14284, (2025)(jul) PDF

3. Gayathri, V., Bartos, I., Rosswog, S., Miller, M., Veske, D., et al. **(Lu, W.)**, Do gravitational wave observations in the lower mass gap favor a hierarchical triple origin?, *submitted*, arXiv: 2307.09097, (2023)(jul) PDF

2. Kulkarni, S., Harrison, F., Grefenstette, B., Earnshaw, H., Andreoni, I., et al. **(Lu, W.)**, Science with the Ultraviolet Explorer (UVEX), *submitted*, arXiv: 2111.15608, (2021)(nov) PDF

1. **Lu, W.**, Beniamini, P., McDowell, A., Deceleration of relativistic jets with lateral expansion, *submitted*, arXiv: 2005.10313, (2020)(may) PDF